Generative AI Analysis and Ethics Report: Simple GAN for MNIST Digit Generation

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GitHub Repository: https://github.com/sandipdey2/generative-mnist

Overview

This project implements a Generative Adversarial Network (GAN) to generate handwritten digit images from the public MNIST dataset. A simple fully connected GAN was trained for 40 epochs on Apple MPS to produce 28×28 grayscale digit images. The goal was a correct, stable implementation that demonstrates the adversarial training process rather than photorealistic output. All reported metrics are taken directly from the accompanying generative_model.ipynb run.

Dataset or Prompt Description

MNIST contains 60,000 training images of real handwritten digits (0–9). Each image is a 28×28 grayscale matrix. The dataset was loaded via torchvision and is publicly available and appropriate for academic use. It supports image generation because the data distribution is well-defined and relatively low-dimensional, allowing a basic GAN to learn digit structure within a 40-epoch training run. No synthetic or AI-generated data were used.

Model Design and Training Approach

The generator maps a 100-dimensional Gaussian latent vector through three fully connected layers (256, 512, 1024) with LeakyReLU activations and a final Tanh layer to produce a 28×28 image. The discriminator is a fully connected network that outputs a probability of real versus fake. Both networks were optimized with Adam (learning rate 0.0002, betas 0.5 and 0.999) and binary cross-entropy loss, a standard configuration for simple GANs (Goodfellow et al., 2014). Training ran for 40 epochs on the full 60,000-image MNIST training set using the MPS device. Generator and discriminator losses were recorded each epoch to monitor the relative strength of the two networks.

Output Evaluation and Interpretation

After 40 epochs the final losses were Discriminator Loss 1.4609 and Generator Loss 1.7989. The generated samples show recognizable digit shapes, especially for simpler digits such as 0, 1, and 7. Many outputs remain blurry or contain incomplete strokes and mixed digit features. A concrete failure case is residual mode collapse: the generator continues to favor a limited set of common digit styles while other modes remain underrepresented. The persistent gap between generator and discriminator losses indicates that the adversarial balance had not fully stabilized even after 40 epochs, which is consistent with the known training dynamics of fully connected GANs (Goodfellow et al., 2014).

Ethical Considerations and Responsible Use

Even a simple digit generator raises questions of creative ownership and potential misuse. If a similar architecture were trained on real faces or medical images without consent, the generated outputs could enable deepfakes or privacy violations (Chesney & Citron, 2019). In this educational setting the training data are public handwritten digits, so the immediate risk is low; however, the same training procedure could be repurposed on more sensitive data. Responsible use therefore requires clear disclosure that outputs are synthetic, avoidance of training on private or copyrighted material without permission, and consideration of downstream applications before deployment.

Limitations and Future Improvements

The fully connected architecture and modest capacity limit visual sharpness. Mode collapse remains visible after 40 epochs, and the final losses (D 1.4609, G 1.7989) show that equilibrium was not reached. Realistic next steps

include switching to a DCGAN-style convolutional architecture, applying spectral normalization or gradient penalty for more stable training, increasing model capacity, and evaluating sample quality with quantitative metrics such as FID. These changes would address the observed blurriness and mode collapse while preserving the same adversarial training framework.

References

Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). Generative adversarial nets. Advances in Neural Information Processing Systems, 27.

Chesney, R., & Citron, D. (2019). Deep fakes: A looming challenge for privacy, democracy, and national security. California Law Review, 107(6), 1753–1820.

PyTorch Documentation. (n.d.). torch.nn and torchvision.datasets.MNIST. https://pytorch.org/docs/